T.C.

Özyeğin University

CS.423-

Homework Assignment 1

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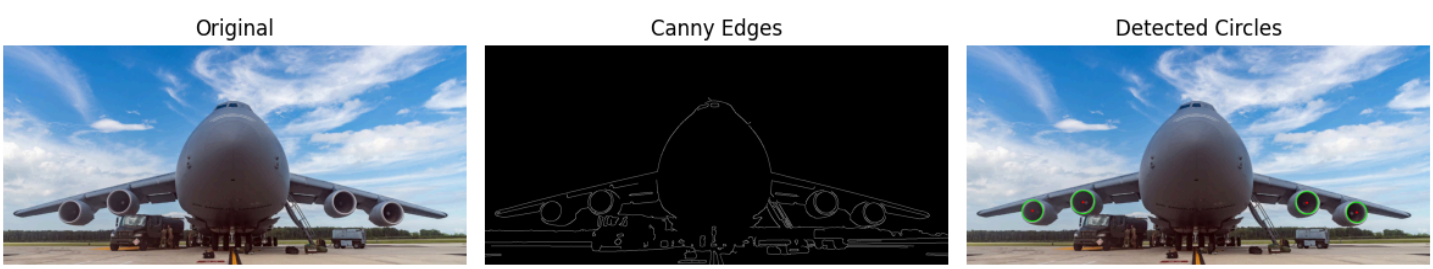
1. Before beginning the assignment, needed libraries are imported, and the drive is mounted.
2. Canny image is used for the circle functions.
3. The dimensions of the image are obtained so circles are prevented from going outside the image boundaries.
4. The number of possible radii is calculated and a 3D accumulator array is created to store votes for every possible circle center (x, y) and radius.
5. All edge pixel coordinates are extracted.
6. Angles from 0 to 360 degrees are generated so that every possible direction around a circle can be considered.
7. For each edge pixel and each radius, possible circle centers are calculated using the circle equation.
8. Invalid coordinates are filtered out so that votes are only given to centers that lie inside the image.
9. The accumulator is updated by adding votes to the corresponding center and radius positions. If a point has many votes, it is more likely to be a real circle.
10. Accumulator cells whose votes exceed a threshold are selected.
11. The candidates are sorted by vote count so the most likely circles are considered first.
12. Circles that are too close to previously selected circles are ignored to avoid detecting the same circle multiple times.
13. The top circles (parameter of hough_circles_peaks called “s”) are kept as the final detected circles.
14. For drawing circles: [OpenCV: Hough Circle Transform]() is referenced

Output:



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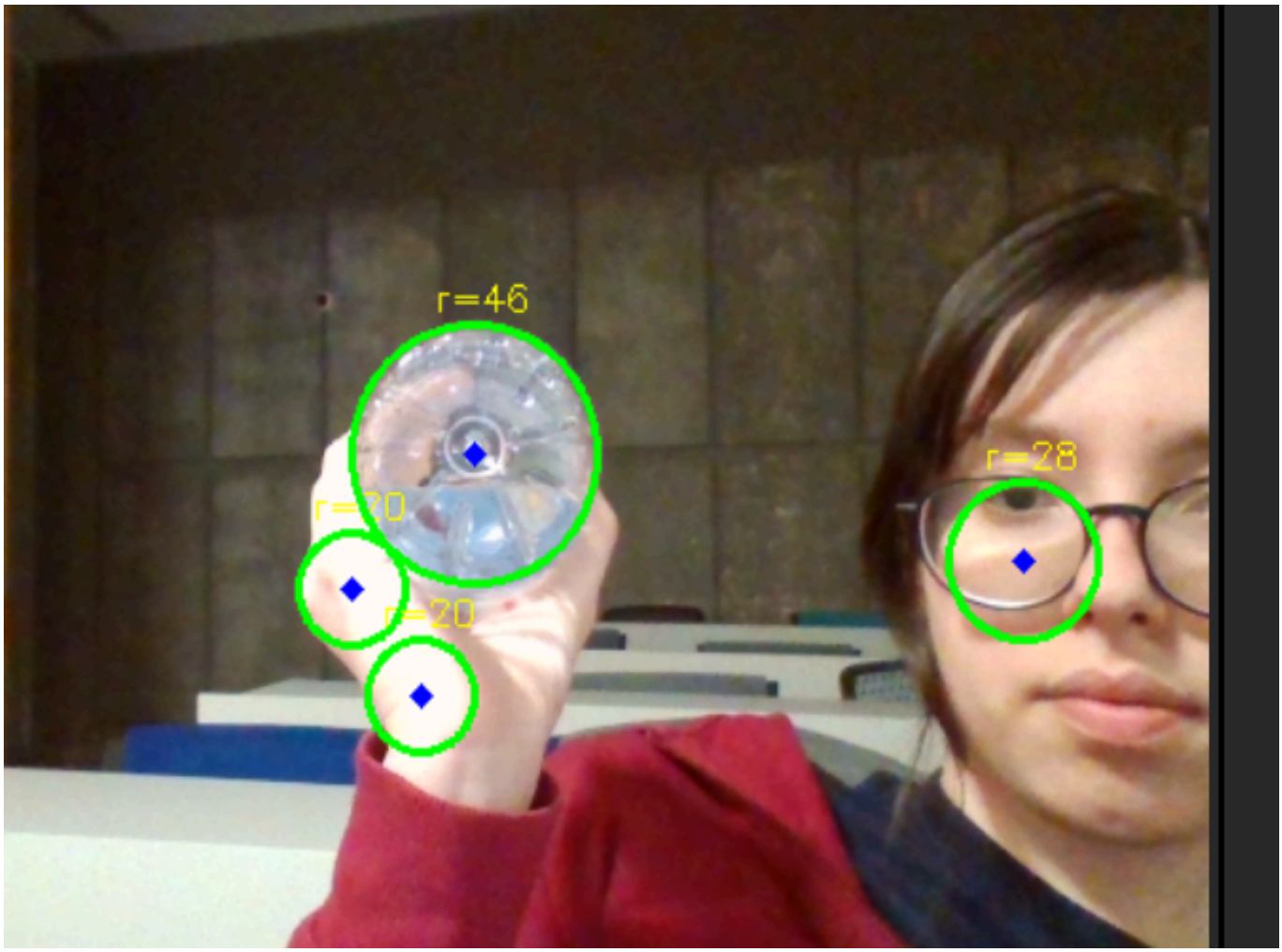
1. **For webcam streaming:** **[colab_webcam.ipynb - Colab]()** is taken as a reference
2. Circle functions are copied from the previous part.
3. Each iteration reads a frame from the webcam, allowing the algorithm to process video frame-by-frame.
4. The webcam frame is decoded into an image format so it can be processed with OpenCV.
5. The image is resized using a scale factor to reduce its resolution, which makes the circle detection faster.
6. The image is converted to grayscale because edge detection works more effectively on single-channel intensity images.

7. The minimum and maximum circle radii are scaled according to the resized image so that the detection still corresponds to the original circle sizes.
8. The Hough circle accumulator is computed using the edge image and the possible radius range, allowing edge pixels to vote for possible circle centers.
9. Detected circles are rescaled back to the original image size so they match the coordinates of the original webcam frame.
10. An overlay image is created containing the detected circles, which will be drawn on top of the video frame.
11. The display label is updated to show the current frame number, how many circles were detected, the radius range, and the vote threshold.
12. The frame counter is increased so the system keeps track of how many frames have been processed.

Output: